

Technical Communication Assignment-8

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April 2026

Question 1

(a) Recursion Tree

The recursion tree for `fun(arr, 0, 7)` is:

- `fun(arr, 0, 7)` [mid = 3]
 - `fun(arr, 0, 3)` [mid = 1]
 - * `fun(arr, 0, 1)` [mid = 0]
 - `fun(arr, 0, 0)` → Base case
 - `fun(arr, 1, 1)` → Base case
 - * `fun(arr, 2, 3)` [mid = 2]
 - `fun(arr, 2, 2)` → Base case
 - `fun(arr, 3, 3)` → Base case
 - `fun(arr, 4, 7)` [mid = 5]
 - * `fun(arr, 4, 5)` [mid = 4]
 - `fun(arr, 4, 4)` → Base case
 - `fun(arr, 5, 5)` → Base case
 - * `fun(arr, 6, 7)` [mid = 6]
 - `fun(arr, 6, 6)` → Base case
 - `fun(arr, 7, 7)` → Base case

(b) Intermediate Arrays

The intermediate arrays after each call:

{2, 1, 3, 4, 5, 6, 7, 8}

{2, 1, 4, 3, 5, 6, 7, 8}

{4, 3, 2, 1, 5, 6, 7, 8}

{4, 3, 2, 1, 6, 5, 7, 8}

{4, 3, 2, 1, 6, 5, 8, 7}

{4, 3, 2, 1, 8, 7, 6, 5}

{8, 7, 6, 5, 4, 3, 2, 1}

(c) Final Array

The final array is:

$\{8, 7, 6, 5, 4, 3, 2, 1\}$

(d) Stack Depth

The maximum recursion depth is:

4

(e) Total Recursive Calls

The total number of recursion calls is:

15

Question 2

(a) Differentiation

$$f(x) = \frac{x^3 - 3x + 2}{\sqrt{x^2 + 1}}$$

Using quotient rule:

$$f'(x) = \frac{u'v - uv'}{v^2}$$

$$u' = 3x^2 - 3, \quad v' = \frac{x}{\sqrt{x^2 + 1}}$$

$$f'(x) = \frac{(3x^2 - 3)\sqrt{x^2 + 1} - (x^3 - 3x + 2)\frac{x}{\sqrt{x^2 + 1}}}{x^2 + 1}$$

$$f'(x) = \frac{2x^4 + 3x^2 - 2x - 3}{(x^2 + 1)^{3/2}}$$

(b) Integration

$$\int_1^e x \ln x \, dx$$

Solution:

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \frac{x^2}{4} + C$$

$$\begin{aligned} \int_1^e x \ln x \, dx &= \left[\frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_1^e \\ &= \frac{e^2}{2} - \frac{e^2}{4} + \frac{1}{4} \end{aligned}$$

$$\int_1^e x \ln x \, dx = \frac{e^2 + 1}{4}$$

(c) Algebra

$$2x^2 - 5x + 3 = 0$$

Solution:

$$(2x - 3)(x - 1) = 0$$

$$x = \frac{3}{2}, \quad x = 1$$

Using quadratic formula:

$$x = \frac{5 \pm \sqrt{25 - 24}}{4}$$

$$x = \frac{3}{2}, \quad x = 1$$

Question 3

(a) Determinant and Inverse of Matrix

Given:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

Step 1: Determinant

$$\begin{aligned} \det(A) &= 2 \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} + 0 \\ &= 2(3 \cdot 1 - 2 \cdot 2) - 1(1 \cdot 1 - 2 \cdot 0) = 2(-1) - 1 = -3 \end{aligned}$$

$$\det(A) = -3$$

Step 2: Cofactors (example)

$$C_{11} = -1, \quad C_{12} = -1, \quad C_{13} = 2$$

Step 3: Inverse

$$A^{-1} = \frac{1}{-3} \text{adj}(A)$$

$$A^{-1} = -\frac{1}{3} \begin{pmatrix} -1 & -1 & 2 \\ -1 & 2 & -4 \\ 2 & -4 & 5 \end{pmatrix}$$

(b) System of Equations (Cramer's Rule)

$$2x + y = 5$$

$$3x - 2y = 4$$

Solution

$$D = -7, \quad D_x = -14, \quad D_y = -7$$

$$x = \frac{D_x}{D} = 2, \quad y = \frac{D_y}{D} = 1$$

$$x = 2, \quad y = 1$$

(c) Vector Operations

$$\vec{u} = (1, 2, 3), \quad \vec{v} = (4, 5, 6)$$

Solution

$$\vec{u} \cdot \vec{v} = 32$$

$$\vec{u} \times \vec{v} = (-3, 6, -3)$$

$$\theta \approx 12.93^\circ$$

Final results: Dot = 32, Cross = $(-3, 6, -3)$, Angle $\approx 12.93^\circ$

(d) Eigenvalues and Eigenvectors

$$B = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

Solution

$$\lambda^2 - 7\lambda + 10 = 0 \implies \lambda = 5, 2$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \vec{v}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

Eigenvalues: 5, 2

Eigenvectors: $\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix}$